

MotorAudit

Technical Documentation

Energy Audit Platform for Industrial Electric Motors

MotorAudit Project

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Abstract

MotorAudit is a web-based engineering platform that helps plant engineers and facility managers quantify the energy-saving potential of three common industrial interventions: replacing an inefficient motor, adding a variable frequency drive (VFD) to a throttled pump or fan, and diagnosing power-quality problems from voltage/current waveforms. The three modules share a common IEC motor efficiency database and can be consolidated into a single PDF audit report. This document describes the system architecture, the engineering formulas implemented in each module, the REST API, and the assumptions and limitations that apply to the results.

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1 Introduction

MotorAudit combines three engineering modules under a single user interface:

1. **Motor Replacement & Payback** — estimates the annual energy, cost, and CO₂ savings from replacing an existing motor with a higher-efficiency model, and the resulting simple payback period.
2. **VFD Sizing & Savings** — estimates the energy savings from installing a variable frequency drive on a throttled pump, fan, or conveyor, based on a user-supplied speed/time load profile.
3. **Power Quality Analyzer** — computes total harmonic distortion (THD), power factor, and phase unbalance from voltage/current waveform samples (uploaded as CSV, submitted as JSON, or generated from built-in presets), and flags out-of-limit conditions with corrective recommendations.

All three modules can optionally be enriched with natural-language engineering commentary generated by a large language model (Google Gemini), and the results of all three can be merged into a single downloadable PDF audit report.

Results throughout the application are explicitly labelled as **engineering estimates** rather than certified measurements; see Section 12 for the full list of assumptions and limitations.

2 System Architecture

MotorAudit is a two-tier web application: a Python backend that performs all engineering calculations and a TypeScript/React frontend that collects inputs and visualizes results.

2.1 Technology Stack

Layer	Technology
Backend framework	FastAPI (Python), served by Uvicorn
Data validation	Pydantic / <code>pydantic-settings</code>
Numerical computation	NumPy
PDF generation	ReportLab
AI insights	Google Gemini (<code>gemini-3-flash-preview</code>)
Frontend framework	Next.js (App Router) / React / TypeScript
Styling	Tailwind CSS
Testing	pytest

Table 1: Core technology stack.

2.2 Repository Layout

```

motoraudit/
|-- backend/
|   |-- app/
|   |   |-- main.py           # FastAPI app entry point
|   |   |-- config.py        # Settings (pydantic-settings)
|   |   |-- db/seed.py       # IEC motor efficiency table
|   |   |-- schemas/         # Pydantic request/response models
|   |   |-- core/
|   |   |   |-- motor_db.py  # Efficiency interpolation

```

```

| | | |-- replacement_calc.py
| | | |-- vfd_calc.py
| | | |-- dsp.py           # FFT, THD, PF, unbalance
| | | |-- gemini.py        # AI insight generation
| | | '-- report_gen.py    # ReportLab PDF generator
| | |-- api/              # FastAPI routers
| | '-- tests/            # pytest unit tests
| |-- requirements.txt
| '-- .env.example
|-- frontend/
| |-- src/
| | |-- app/              # Next.js App Router pages
| | |-- components/       # Per-module React components + charts
| | |-- lib/              # TypeScript types, API client
| | '-- hooks/            # useAuditSession context
| '-- package.json
'-- README.md

```

2.3 Request Flow

Each module follows the same pattern: the frontend collects structured input through a form component, POSTs it to the corresponding FastAPI endpoint, the backend validates it against a Pydantic schema, dispatches to a pure calculation function in `app/core/`, and returns a structured JSON response that the frontend renders as tables and charts (Recharts). The optional AI-insights endpoint and the PDF report endpoint both consume the same calculation result objects, ensuring the report and the on-screen results are always consistent.

2.4 Configuration

Backend configuration is loaded from a `.env` file via `pydantic-settings` (`app/config.py`). Key settings include:

Setting	Default	Purpose
<code>cors_origins</code>	<code>localhost:3000</code>	Allowed frontend origin(s)
<code>default_tariff_per_kwh</code>	<code>0.12</code>	Default electricity tariff
<code>default_co2_factor_kg_per_kwh</code>	<code>0.233</code>	Grid CO ₂ intensity
<code>default_lifetime_years</code>	<code>20</code>	Motor lifetime for savings analysis
<code>default_vfd_efficiency</code>	<code>0.97</code>	Default VFD drive efficiency
<code>default_vfd_headroom</code>	<code>1.10</code>	VFD sizing headroom factor
<code>pq_thd_voltage_limit_pct</code>	<code>5.0</code>	Voltage THD status-flag limit
<code>pq_thd_current_limit_pct</code>	<code>8.0</code>	Current THD status-flag limit
<code>pq_power_factor_min</code>	<code>0.95</code>	Power-factor status-flag limit
<code>pq_current_unbalance_pct</code>	<code>3.0</code>	Current-unbalance status-flag limit
<code>gemini_api_key</code>	<code>(empty)</code>	Enables AI-insights endpoints

Table 2: Selected backend configuration settings.

3 Installation and Quick Start

3.1 Backend

```
cd motoraudit/backend
python3 -m venv .venv && source .venv/bin/activate    # Windows: .venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env
uvicorn app.main:app --reload --port 8000
# API docs -> http://localhost:8000/docs
```

3.2 Frontend

```
cd motoraudit/frontend
npm install
cp .env.example .env.local
npm run dev
# App -> http://localhost:3000
```

3.3 Tests

```
cd motoraudit/backend
pytest -v
```

Unit tests cover the DSP engine (`test_dsp.py`), the replacement calculator (`test_replacement.py`), and the VFD calculator (`test_vfd.py`).

4 Module 1: Motor Replacement & Payback

4.1 Engineering Rationale

A motor that runs 6000 h/yr at 75% load for 20 years consumes roughly ten times its own purchase price in electricity. Moving from an IE1 to an IE3 efficiency class typically saves 2–4% of input energy at the operating load point; on large machines running long hours, this translates into substantial annual savings.

4.2 Motor Efficiency Model

The IEC 60034-30-1:2014 standard (Table 1) tabulates minimum guaranteed efficiency η at four standard load points: 25%, 50%, 75%, and 100% of rated load, for each IE class (IE1–IE4), motor pole count, and rated power. The efficiency at an arbitrary load factor LF is obtained by fitting a least-squares quadratic polynomial through the four tabulated (LF, η) points and evaluating it at the requested load factor:

$$\eta(\text{LF}) = a \text{LF}^2 + b \text{LF} + c \quad (1)$$

where a , b , c are obtained from a quadratic least-squares fit through the four IEC points $\text{LF} \in \{0.25, 0.50, 0.75, 1.00\}$. A quadratic is adequate because the η -vs-LF curve of an induction motor is smooth and unimodal over this range. The result is clipped to $[0.30, 1.00]$ as a numerical safeguard.

If the user instead supplies a single full-load efficiency value (`efficiency_override_pct`) rather than an IE class, a parametric Gaussian-peak model is used instead of the IEC table:

$$\eta(\text{LF}) = \eta_{\text{peak}} \cdot \exp(-k (\text{LF} - \text{LF}_{\text{peak}})^2) \quad (2)$$

with $LF_{\text{peak}} = 0.75$ and k chosen so that $\eta(1.0) \approx 0.985 \eta_{\text{peak}}$, i.e. $k = -\ln(0.985)/0.0625 \approx 0.243$. The requested load factor is clamped to $[0.05, 1.05]$ before evaluation in both models.

Data source. IEC 60034-30-1:2014 Table 1 minimum efficiency values, 4-pole 50 Hz motors, IE1–IE4, 4–110 kW. These are minimum *guaranteed* values — name-plate efficiencies of specific commercial models are typically 1–2% higher.

4.3 Energy and Cost Calculations

Electrical input power at the operating load factor is the shaft output power divided by efficiency:

$$P_{\text{shaft}} = P_{\text{rated}} \cdot LF, \quad P_{\text{in}} = \frac{P_{\text{shaft}}}{\eta(LF)} \quad (3)$$

Annual energy consumption:

$$E_{\text{annual}} = \frac{P_{\text{rated}} \cdot LF}{\eta(LF)} \cdot H \quad [\text{kWh/yr}] \quad (4)$$

where H is the annual operating hours. The same load factor and rated power are used for both the existing and replacement motor, so that only the efficiency term changes between the two scenarios.

Annual savings from replacing the existing motor (E_{old}) with the new one (E_{new}):

$$\Delta E = E_{\text{old}} - E_{\text{new}} \quad [\text{kWh/yr}] \quad (5)$$

$$\Delta \text{Cost} = \Delta E \cdot \text{tariff} \quad [\text{currency/yr}] \quad (6)$$

$$\Delta \text{CO}_2 = \Delta E \cdot \text{co2_factor} \quad [\text{kg CO}_2/\text{yr}] \quad (7)$$

Simple (undiscounted) payback period:

$$\text{Payback} = \frac{C_{\text{motor}}}{\Delta \text{Cost}} \quad [\text{years}] \quad (8)$$

Lifetime net savings over the assumed motor lifetime L (also undiscounted, i.e. no NPV/IRR):

$$\text{LNS} = \Delta \text{Cost} \cdot L - C_{\text{motor}} \quad (9)$$

4.4 API Contract Summary

Request (ReplacementRequest): `existing_motor` and `replacement_motor` objects (each specifying either `ie_class` or `efficiency_override_pct`, plus `rated_power_kw` and `load_factor` for the existing motor, and `cost` for the replacement), `operating_hours_per_year`, `tariff_per_kwh`, optional `co2_factor_kg_per_kwh`, and `lifetime_years`.

Response (ReplacementResponse): per-motor operating points (`existing`, `replacement`), a savings summary, an `efficiency_curve` for charting η vs. load factor for both motors, a `payback_timeline` (cumulative savings and net position by year), and a list of human-readable assumptions describing the exact inputs used.

5 Module 2: VFD Sizing & Savings

5.1 Engineering Rationale

Pump and fan systems are among the largest categories of industrial electricity use. When flow is controlled by throttling a valve or damping a fan, the motor continues to run near full speed and the excess energy is wasted across the restriction. A variable frequency drive (VFD) instead reduces motor speed to match actual flow demand. Because shaft power on centrifugal loads follows the **affinity law** (power $\propto$ speed³), running at 70% speed consumes only $0.7^3 \approx 34\%$ of full-speed power.

5.2 Load Models

Centrifugal loads (pump / fan) obey the cubic affinity law:

$$P_{\text{shaft}}(n) = P_{\text{rated}} \cdot \left(\frac{n}{n_{\text{rated}}} \right)^3 \quad (10)$$

Constant-torque loads (conveyor) scale linearly with speed:

$$P_{\text{shaft}}(n) = P_{\text{rated}} \cdot \frac{n}{n_{\text{rated}}} \quad (11)$$

In both cases, the electrical input through the VFD-driven motor is:

$$P_{\text{elec}} = \frac{P_{\text{shaft}}}{\eta_{\text{motor}} \cdot \eta_{\text{VFD}}} \quad (12)$$

Throttling baseline. In the absence of a VFD, the motor is modelled as drawing a constant electrical input of $P_{\text{rated}}/\eta_{\text{motor}}$ at every operating point, regardless of actual flow demand. This is a deliberately conservative assumption (some motors draw marginally less at high slip) and is the standard convention for VFD savings calculations.

5.3 Annual Energy over a Load Profile

The user supplies a load profile as a list of (n_i, t_i) pairs, where n_i is the speed fraction and t_i is the fraction of annual operating hours spent at that speed ($\sum_i t_i = 1$, enforced within ± 0.02 tolerance). For each profile point i :

$$E_i = P_i \cdot t_i \cdot H \quad [\text{kWh/yr}] \quad (13)$$

Total baseline and VFD energy are the sums of E_i over all profile points using the respective P_i model above. Savings and payback follow the same form as Module 1:

$$\Delta E = E_{\text{baseline}} - E_{\text{VFD}}, \quad \text{Payback} = \frac{C_{\text{VFD}}}{\Delta E \cdot \text{tariff}} \quad (14)$$

5.4 VFD Sizing

The recommended VFD power rating includes a headroom margin to cover starting-torque boost and thermal derating:

$$P_{\text{VFD}} = P_{\text{motor}} \cdot \text{headroom_factor} \quad (15)$$

with a default headroom factor of 1.10 (10%).

5.5 API Contract Summary

Request (VFDDRequest): `motor_power_kw`, `load_type` (pump / fan / conveyor), `motor_efficiency`, `vfd_efficiency`, a `load_profile` list of {`speed_fraction`, `time_fraction`} points (1–20 entries, time fractions summing to 1.0), `operating_hours_per_year`, `tariff_per_kwh`, `vfd_cost`, and `headroom_factor`.

Response (VFDResponse): `recommended_vfd_kw`, a per-point `profile_analysis` breakdown, aggregate totals (baseline/VFD annual energy, savings, payback), a `power_vs_speed` curve (10 points from 10% to 100% speed) for charting, and a list of `assumptions`.

6 Module 3: Power Quality Analyzer

6.1 Engineering Rationale

Non-linear loads — VFDs, rectifiers, UPS systems — inject harmonic currents that increase I^2R losses, overheat motors and transformers, and can cause nuisance protective-relay tripping. Voltage unbalance above roughly 2% reduces motor efficiency and shortens motor life. Low power factor increases apparent current draw and, in many tariff structures, incurs utility penalty charges. Module 3 analyzes voltage and current waveforms (three-phase or single-phase) to quantify these effects.

6.2 Signal Processing Pipeline

Each channel (`ia`, `ib`, `ic`, `va`, `vb`, `vc`) is processed with a Hanning-windowed FFT to reduce spectral leakage:

$$X[k] = \text{FFT}(x[n] \cdot w[n]), \quad w[n] = 0.5 \left(1 - \cos \frac{2\pi n}{N-1} \right) \quad (16)$$

Peak amplitude at each frequency bin is corrected for the window's energy loss, and converted to RMS:

$$A_{\text{peak}}[k] = \frac{2|X[k]|}{\sum_n w[n]}, \quad A_{\text{rms}}[k] = \frac{A_{\text{peak}}[k]}{\sqrt{2}} \quad (17)$$

For each harmonic order $h = 1, 2, \dots$, the analyzer searches a ± 3 Hz window around the target frequency $h \cdot f_0$ (to tolerate slight frequency drift between the signal and the sample-rate grid) and takes the maximum RMS amplitude found as I_h (or V_h).

6.3 Total Harmonic Distortion (THD)

Following the IEEE 519-2022 definition:

$$\text{THD} = \frac{\sqrt{\sum_{h=2}^N I_h^2}}{I_1} \times 100\% \quad (18)$$

where I_1 is the RMS amplitude of the fundamental and I_h are the RMS amplitudes of the harmonic components (orders 1–15 are always retained; higher orders are retained only if their amplitude exceeds 0.3% of the fundamental).

6.4 RMS and Peak

$$X_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}, \quad X_{\text{peak}} = \max_i |x_i| \quad (19)$$

6.5 Power Metrics

Given synchronized voltage $v(t)$ and current $i(t)$ samples for one phase:

$$P = \overline{v \cdot i} \quad (\text{active power}) \quad (20)$$

$$S = V_{\text{rms}} \cdot I_{\text{rms}} \quad (\text{apparent power}) \quad (21)$$

$$Q = \sqrt{\max(S^2 - P^2, 0)} \quad (\text{reactive power}) \quad (22)$$

$$\text{PF} = \frac{P}{S} \quad (\text{true power factor}) \quad (23)$$

The **displacement power factor** $\cos \varphi$ is computed from the phase angle between the fundamental voltage and current phasors (obtained from the FFT bin nearest the detected fundamental frequency):

$$\varphi = \angle V_1 - \angle I_1, \quad \cos \varphi = |\cos(\varphi)| \quad (24)$$

Note that $\text{PF} < \cos \varphi$ whenever harmonic distortion is present, since PF (true power factor) includes the effect of distortion power, while $\cos \varphi$ reflects only the fundamental-frequency displacement.

Reactive power correction. When PF falls below the configured target (default 0.95), the required shunt capacitor bank rating to reach the target is:

$$Q_{\text{corr}} = P \cdot (\tan \varphi_{\text{actual}} - \tan \varphi_{\text{target}}) \quad [\text{kVAr}] \quad (25)$$

6.6 Phase Unbalance (NEMA MG-1)

For a three-phase system with RMS phase currents I_a, I_b, I_c :

$$\bar{I} = \frac{I_a + I_b + I_c}{3}, \quad \text{UB}\% = \frac{\max_x |I_x - \bar{I}|}{\bar{I}} \times 100 \quad (26)$$

The same formula is applied to phase voltages when all three voltage channels are present.

6.7 Status Flags and Recommendations

Each computed metric is compared against a configurable threshold (`PQThresholds: thd_voltage_pct`, `thd_current_pct`, `power_factor_min`, `current_unbalance_pct`) and classified as **PASS**, **WARN** (within 20% over the limit, or within 2% under a “low-is-bad” limit such as power factor), or **FAIL**. A rule-based recommendation engine then produces plain-language

corrective guidance — e.g. identifying dominant 5th/7th harmonics as indicative of a 6-pulse VFD or rectifier load and suggesting a passive or active harmonic filter, or sizing a capacitor bank from the computed Q_{corr} .

These thresholds and recommendations are **advisory** engineering estimates approximating IEEE 519-2022 general system limits; they do not constitute legal or regulatory compliance certification.

6.8 Preset Signals

For demonstration and testing without uploaded data, four synthetic waveform presets are available (sampled at 10 kHz, 50 Hz fundamental, 0.2 s duration):

Preset	Description
<code>clean_sinusoid</code>	Pure 50 Hz, 42 A / 230 V — zero-THD reference
<code>vfd_harmonics</code>	6-pulse VFD current: 5th (14%), 7th (9%), 11th (5%), 13th (3%)
<code>unbalanced_3phase</code>	Three-phase with phase B -8% , phase C $+6\%$ unbalance
<code>low_power_factor</code>	Single-phase, 37° lag $\rightarrow$ PF ≈ 0.80

Table 3: Built-in Module 3 waveform presets.

6.9 API Contract Summary

Request (PQRequest): `source` (json or preset), optional `preset_name`, `sample_rate_hz`, `fundamental_freq_hz`, `thresholds`, and optional `data` (a `WaveformData` object with `ia` required and `ib`, `ic`, `va`, `vb`, `vc` optional, each a list of ≥ 64 samples of equal length). A dedicated upload endpoint additionally accepts the waveform as a CSV file with columns `time`, `ia`, `ib`, `ic`, `va`, `vb`, `vc`].

Response (PQResponse): `duration_s`, `actual_fundamental_hz` (detected from the `ia` or `va` FFT, not assumed), per-channel `channels` analysis (RMS, peak, THD, harmonic table), optional `power_metrics`, optional `phase_unbalance`, a `status_table` of `StatusFlag` entries, a list of `recommendations`, and downsampled `waveform_preview` and `fft_preview` data (log-magnitude in dBFS, 0–2 kHz) for charting.

7 AI-Generated Engineering Insights

Each module has a corresponding `/api/ai/*` endpoint that sends the structured calculation result to Google’s Gemini model (`gemini-3-flash-preview`) with a role-primed prompt (“a senior energy engineer with 20 years of experience”) and a fixed three-paragraph output template tailored to that module:

- **Replacement:** business-case framing (is the payback attractive?), sensitivity analysis (which input has the most leverage?), and pre-commissioning checks.
- **VFD:** explanation of why the cubic affinity-law savings are physically expected, commonly overlooked installation risks (PWM dV/dt insulation stress, minimum-speed cooling limits, harmonic injection, bearing currents), and further optimization strategies.
- **Power Quality:** likely equipment source of the observed harmonic signature, quantified consequences for connected equipment and billing, and a prioritized corrective sequence.

The Gemini client is lazily initialized; if `GEMINI_API_KEY` is not configured, the AI-insights endpoints return 503 **Service Unavailable** while the rest of the application continues to function normally.

8 PDF Report Generation

The `POST /api/report/summary` endpoint accepts the results of any combination of the three modules and returns a consolidated JSON summary. The `POST /api/report/pdf` endpoint performs the same consolidation and streams a formatted PDF audit report, generated with ReportLab, suitable for direct delivery to a client or inclusion in a plant energy-audit file.

9 REST API Reference

All endpoints are served under the `/api` prefix. Interactive Swagger UI documentation is available at <http://localhost:8000/docs> (ReDoc at `/redoc`) whenever the backend is running.

Method	Path	Description
GET	<code>/api/health</code>	Service health check
GET	<code>/api/motors</code>	List seeded IEC motors
GET	<code>/api/replacement/demo</code>	Demo preset inputs for Module 1
POST	<code>/api/replacement/calculate</code>	Run Module 1 calculation
GET	<code>/api/vfd/demo</code>	Demo preset inputs for Module 2
POST	<code>/api/vfd/calculate</code>	Run Module 2 calculation
GET	<code>/api/power-quality/presets</code>	List available Module 3 preset names
POST	<code>/api/power-quality/analyze</code>	Run Module 3 analysis on JSON waveform data
POST	<code>/api/power-quality/upload</code>	Run Module 3 analysis on an uploaded CSV file
POST	<code>/api/report/summary</code>	Consolidated JSON summary across modules
POST	<code>/api/report/pdf</code>	Consolidated PDF report download
POST	<code>/api/ai/replacement</code>	AI insights for a Module 1 result
POST	<code>/api/ai/vfd</code>	AI insights for a Module 2 result
POST	<code>/api/ai/power-quality</code>	AI insights for a Module 3 result

Table 4: MotorAudit REST API endpoints.

10 Frontend Application

The frontend is a Next.js (App Router) application with one route per module: `/replacement`, `/vfd`, `/power-quality`, plus a `/simulation` route hosting an interactive motor simulation (`MotorSimulation.tsx`). Each module route pairs an input form component (e.g. `ReplacementForm.tsx`, `VFDForm.tsx`, `PQForm.tsx`) with a results component (e.g. `ReplacementResults.tsx`) and one or more Recharts-based visualizations (e.g. `EfficiencyVsLoadChart.tsx`, `PaybackTimelineChart.tsx`, `PowerVsSpeedChart.tsx`, `FFTSpectrumChart.tsx`, `WaveformChart.tsx`). Shared components include `FormulaBox.tsx` (renders the governing formula alongside results), `AIInsights.tsx` (displays the Gemini-generated commentary), and `AuditSummaryPanel.tsx`. Cross-module state — allowing a user to run all three modules and then generate one consolidated report — is held in the `useAuditSession` React context (`hooks/useAuditSession.ts`), and all HTTP calls to the backend are centralized in `lib/api.ts` with request/response types declared in `lib/types.ts`.

11 Example Results (Demo Presets)

11.1 Module 1 — 37 kW IE1 → IE3, 75% Load, 6 000 h/yr, \$0.12/kWh

- Annual energy saved: $\approx$ **8 280 kWh**
- Annual cost saved: $\approx$ **\$994**
- Simple payback: $\approx$ **4.2 years** (at \$4 200 motor cost)
- CO₂ reduction: $\approx$ **1 929 kg/yr** (at 0.233 kg/kWh)

11.2 Module 2 — 55 kW Pump VFD, Profile 10/30/40/20% at 100/85/70/50% Speed, 7 000 h/yr

- Baseline: $\approx$ **415 800 kWh/yr** (throttling)
- VFD: $\approx$ **246 000 kWh/yr**
- Saved: $\approx$ **170 000 kWh/yr** ($\approx$ 41% reduction)
- Payback: $\approx$ **0.4 years** at \$7 500 VFD cost

11.3 Module 3 — VFD Harmonics Preset

- Current THD: $\approx$ **18%** → **FAIL** ($>$ 8% limit)
- Voltage THD: $\approx$ **1%** → **PASS**
- Power factor: **0.999** → **PASS** (clean voltage, no reactive component in demo)
- Recommendation: passive harmonic filter targeting the 5th and 7th harmonics

12 Assumptions and Limitations

- Motor efficiency data is sourced from IEC 60034-30-1:2014 minimum guaranteed values (4-pole, 50 Hz). Real motors are typically 1–2% more efficient than the tabulated minimum.
- Efficiency interpolation uses a quadratic polynomial fit through four IEC load points; accuracy degrades for load factors outside the 25–100% range.
- VFD savings assume a quadratic (resistance-dominated) system curve. A system with a significant static head component will show lower actual savings than predicted.
- Power-quality thresholds are advisory approximations of IEEE 519-2022 general limits. No legal or regulatory compliance certification is implied by PASS/WARN/FAIL status flags.
- All financial calculations are undiscounted — no net present value (NPV) or internal rate of return (IRR) is computed. Users requiring rigorous investment analysis should apply an appropriate discount rate to the reported cash flows.
- Results are labelled as **engineering estimates** throughout the user interface and generated reports.

13 References

1. IEC 60034-30-1:2014, *Rotating electrical machines — Part 30-1: Efficiency classes of line operated AC motors (IE code)*.
2. IEEE Std 519-2022, *IEEE Standard for Harmonic Control in Electric Power Systems*.
3. NEMA MG-1, *Motors and Generators*, National Electrical Manufacturers Association.